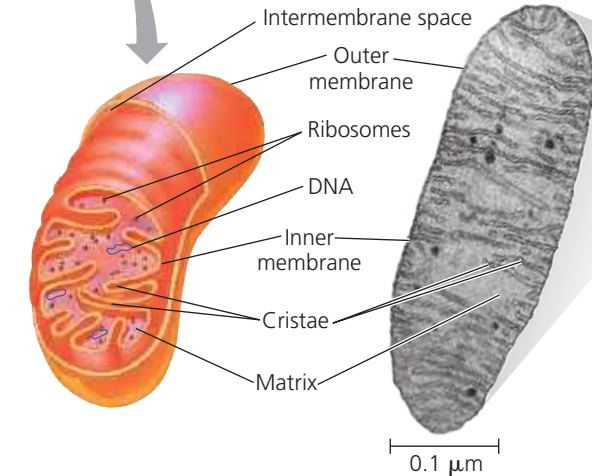
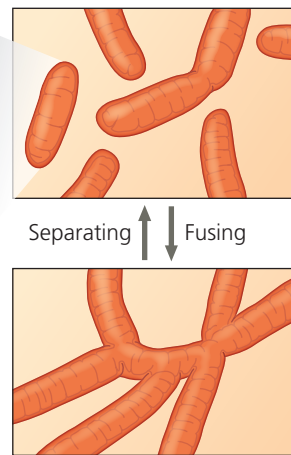


▼ **Figure 6.17 The mitochondrion, site of cellular respiration.** (a) The inner and outer membranes seen in the drawing and TEM establish two compartments: the intermembrane space and the mitochondrial matrix. The cristae increase the surface area of the inner membrane. Circular DNA molecules are associated with the inner mitochondrial membrane. (b) Mitochondria are dynamic, at times separating into fragments or fusing. (c) The LM shows one protist (*Euglena gracilis*) at a much lower magnification than the TEM. The mitochondria form a branched tubular network; the mitochondrial matrix is stained green. The nuclear DNA is stained red; molecules of mitochondrial DNA appear as bright yellow spots.

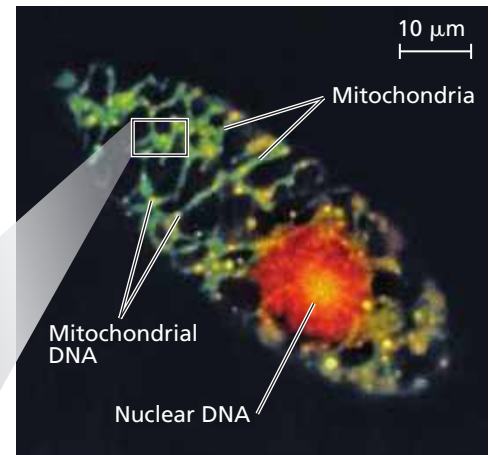
➔ Mastering Biology BioFlix® Animation: Mitochondria



(a) Diagram and TEM of mitochondrion



(b) Dynamic nature of mitochondrial networks



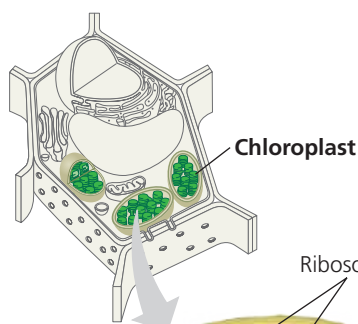
(c) Network of mitochondria in *Euglena* (LM)

during photosynthesis. (You will learn more about photosynthesis in Chapter 10.)

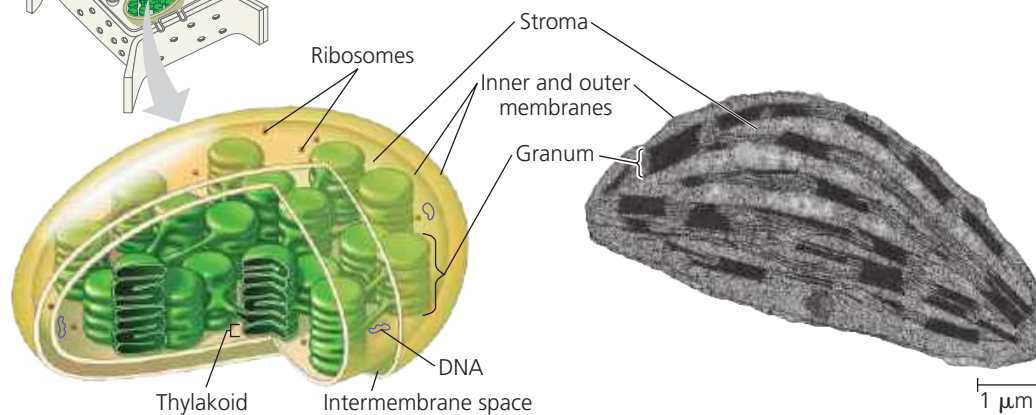
As with mitochondria, the static and rigid appearance of chloroplasts in micrographs or schematic diagrams cannot accurately depict their dynamic behavior in the living cell. Their shape is changeable, and they grow and occasionally pinch in two, reproducing themselves. They are mobile and, with mitochondria and other organelles, move around the

cell along tracks of the cytoskeleton, a structural network we will consider in Concept 6.6.

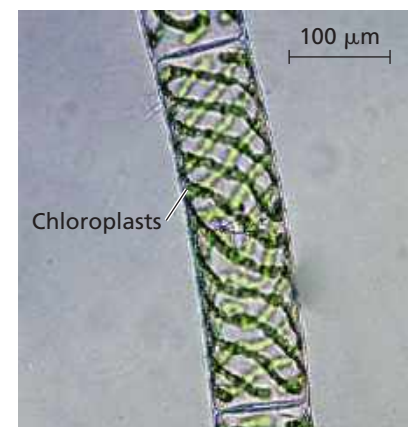
The chloroplast is a specialized member of a family of closely related plant organelles called **plastids**. One type of plastid, the *amyloplast*, is a colorless organelle that stores starch (amylose), particularly in roots and tubers. Another is the *chromoplast*, which has pigments that give fruits and flowers their orange and yellow hues.



▼ **Figure 6.18 The chloroplast, site of photosynthesis.** (a) Many plants have lens-shaped chloroplasts, as shown here in a diagram and a TEM. A chloroplast has three compartments: the intermembrane space, the stroma, and the thylakoid space. Free ribosomes are present in the stroma, as are copies of chloroplast DNA molecules. (b) This light micrograph, at a much lower magnification than the TEM, shows a whole cell of the green alga *Spirogyra crassa*, which is named for the spiral arrangement of each cell's chloroplasts.



(a) Diagram and TEM of chloroplast



(b) Chloroplasts in an algal cell

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